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Breaking Report

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Rh-Catalyzed Kinetic Resolution-Based Enantioselective [4+2]-Cycloaddition-Isomerization of Allene-1,3-dienes

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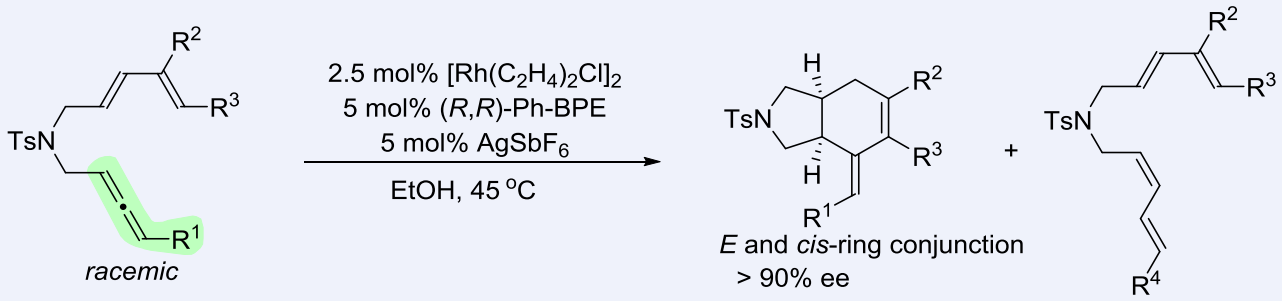
Keywords

Rhodium | Cycloaddition | Kinetic resolution | Allenes | Isomerization

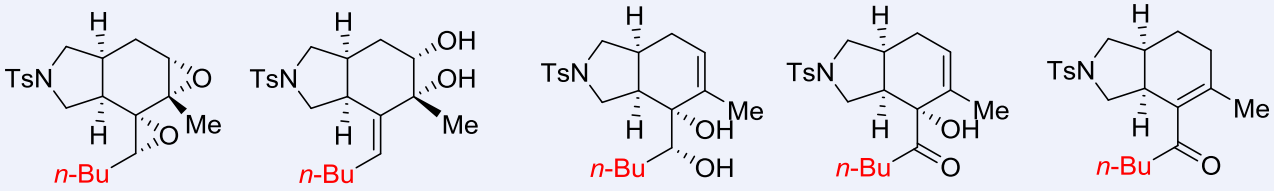
Main observation and conclusion

The first example of Rh-catalyzed kinetic resolution of 1,3-disubstituted allene-1,3-dienes involving intramolecular [4+2]-cycloaddition has been developed. Followed by C=C isomerization, the reaction finally afforded aza-[4.3.0]bicyclic compounds with a very high enantio- and *E*-selectivity. The mechanism has been unveiled by careful control experiments. The synthetic potentials of highly enantioenriched bicyclic products have also been demonstrated.

Comprehensive Graphic Content



- Efficient kinetic resolution
- [4+2]-cycloaddition-C=C bond migration
- Excellent enantioselectivity
- Excellent C=C bond selectivity



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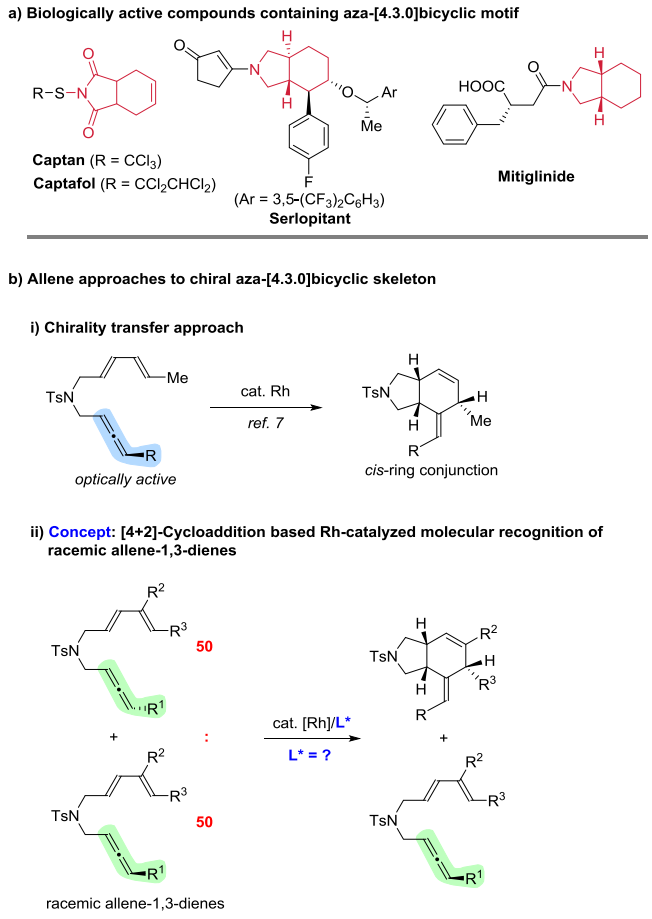
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Background and Originality Content

Aza-[4.3.0]bicyclic structure is a core motif of many biologically active compounds: for example, captan[1a] and captafol[1b] are common agrochemicals for sterilization; serlopitant,[1c] mitiglinide,[1d] and its derivatives have bioactivities for the treatment of prurigo nodulari and type 2 diabetes, respectively (Scheme 1a). Intramolecular [4+2]-cycloadditions between 1,3-dienes and alkenes or alkynes are among the most powerful tools for the construction of such bicyclic functionalized compounds.[2-4] Intramolecular [4+2]-cycloadditions between 1,3-dienes and allenes may also afford aza-[4.3.0]bicyclic products with extra handle for synthetic elaboration.[5] However, there is an issue of axial chirality in the allenes as compared to alkenes and alkynes, and such enantioselective reaction with axially chiral allenes has never been established. Challenges are the recognition of the axial chirality in the starting materials and the enantioselectivity as well as the *Z*/*E*-selectivity for the remaining C=C bond. There are reports on synthesis of such optically active bicyclic products from optically active substrates based on chirality transfer (Scheme 1b-i).[7] We wish to report the first example of Rh-catalyzed sequential [4+2] cycloaddition/C=C bond isomerization of racemic allene-dienes, in which both kinetic resolution of the axially chiral substrates and enantioselective cycloaddition were achieved to afford aza-[4.3.0]-bicyclic compounds in high enantioselectivity (Scheme 1b-ii).

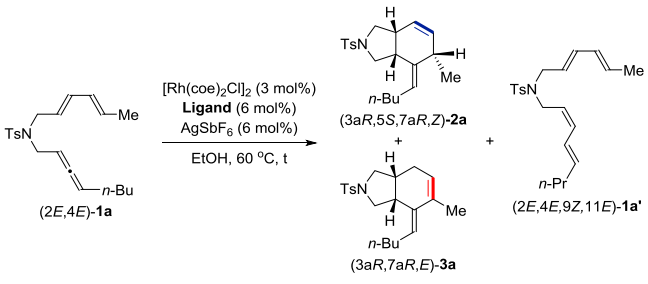
Scheme 1 Typical examples of biologically active aza-[4.3.0]bicyclic compounds and asymmetric intramolecular [4+2]-cycloadditions of nitrogen-tether allene-dienes



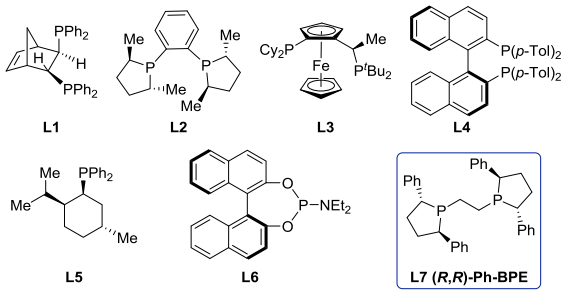
Results and Discussion

Our initial studies began with the [4+2]-cycloaddition of racemic allene-diene (2*E*,4*E*)-**1a** under the catalysis of $[Rh(coe)_2Cl]_2$ and $AgSbF_6$ with different chiral ligands in EtOH at 60 °C (Table 1). (*R*,*R*)-Norphos [(2*R*,3*R*)-(–)-2,3-bis(diphenylphosphino)-bicyclo-[2.2.1]hept-5-ene] **L1** was initially employed as the chiral ligand, and normal [4+2]-cycloaddition product **2a**[7b] was formed in 59% yield with 42% ee with no starting material remained (entry 1). (*R*,*R*)-Me-DUPHOS (1,2-bis[(2*R*,5*R*)-2,5-dimethyl-1-phospholanyl]-benzene) **L2** slightly improved the yield to 64% and ee to 56%. Ferrocene-based phosphine ligand (**L3**) failed to afford **2a** and (*S*)-*p*-Tol-BINAP [(*S*)-2.2'-Bis(di-*p*-tolylphosphino)-1,1'-binaphthyl] **L4** was also inefficient for this reaction (entries 3 and 4). Menthol derived ligand **L5** and Feringa ligand **L6** were also efficient in producing the product **2a**, however, the enantioselectivity is very low (entries 5 and 6). Surprisingly, when (*R*,*R*)-Ph-BPE {1,2-bis[(2*R*,5*R*)-2,5-diphenylphospholan-1-yl]ethane} **L7** was used as the ligand, we did not observe the formation of normal [4+2]-cycloaddition product **2a**. Instead, the sequential [4+2]-cycloaddition and C=C bond isomerization product **3a** was obtained in 29% yield with 95% ee. Instead of recovering starting material, 47% NMR yield of tetra-ene (2*E*,4*E*,9*Z*,11*E*)-**1a'** was also observed (entry 7).

Table 1 Ligand screening[a]



entry	ligand	*t*/h	recovery of (2*E*,4*E*)-**1a**/%	(3a*R*,5*S*,7a*R*,*Z*)-**2a**		(3a*R*,7a*R*,*E*)-**3a**	
				yield[b]/%	ee[c]/%	yield[b]/%	ee[c]/%
1	**L1**	6	0	59	42	0	–
2	**L2**	6	0	64	56	0	–
3	**L3**	19	90	trace	–	0	–
4	**L4**	9	58	trace	–	0	–
5[d]	**L5**	6	0	56	1	0	–
6	**L6**	4	0	79	3	0	–
7[e,f]	**L7**	3	0	0	–	29	-95



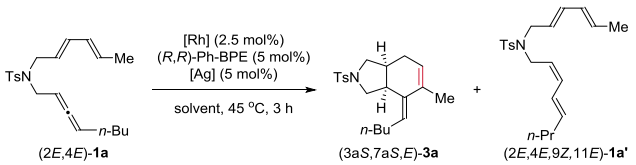
[a] Reaction conditions: 0.1 mmol of (2*E*,4*E*)-**1a**, 3 mol% of $[Rh(coe)_2Cl]_2$, 6 mol% of $AgSbF_6$, and 6 mol% of ligand in EtOH (1.0 mL) at 60 °C. [b] Determined by ^1H NMR analysis using mesitylene as the internal standard. [c] Determined by HPLC analysis. [d] 12 mol% ligand. [e] 0.2 mmol of (2*E*,4*E*)-**1a**, 3 mol% of $[Rh(coe)_2Cl]_2$, 6 mol% of $AgSbF_6$, and 6 mol% of ligand in EtOH (2.0 mL) at 45 °C for 3 h. [f] 47% of (2*E*,4*E*,9*Z*,11*E*)-**1a'** was also observed by ^1H NMR analysis using mesitylene as the internal standard.

After identifying (*R*,*R*)-Ph-BPE as the best ligand, other reaction parameters were examined (Table 2). Firstly, we found the catalyst loading could be reduced to 2.5 mol% without losing the efficiency for the transformation (entry 1). Next, various silver salts were screened, and $AgSbF_6$ was found to be optimal (entries 2—4), which revealed that anions have an effect on the reaction. *n*-PrOH was inappropriate, resulting in 18% of **3a** albeit with 98%

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Table 2 Optimization of reaction conditions[a]



entry	[Rh]	[Ag]	solvent	recovery of (2*E*,4*E*)-**1a**/%	(3a*S*,7a*S*,*E*)-**3a** yield[b]/%	(3a*S*,7a*S*,*E*)-**3a** ee[c]/%	yield of (2*E*,4*E*,9*Z*,11*E*)-**1a'**[b]/%
1	$[Rh(coe)_2Cl]_2$	$AgSbF_6$	EtOH	0	31	>99	45
2[d]	$[Rh(coe)_2Cl]_2$	AgOTf	EtOH	85	0	–	0
3[d]	$[Rh(coe)_2Cl]_2$	$AgBF_4$	EtOH	0	24	>99	44
4[d]	$[Rh(coe)_2Cl]_2$	$AgNTf_2$	EtOH	0	31	98	46
5	$[Rh(coe)_2Cl]_2$	$AgSbF_6$	*n*-PrOH	0	18	98	41
6	$[Rh(coe)_2Cl]_2$	$AgSbF_6$	MeOH	74	0	–	8
7	$[Rh(coe)_2Cl]_2$	$AgSbF_6$	toluene	70	0	–	0
8	[Rh(nbd)Cl]$_2$	$AgSbF_6$	EtOH	57	0	–	7
9	***$[Rh(C_2H_4)_2Cl]_2$***	***$AgSbF_6$***	***EtOH***	***0***	***36***	***>99***	***52***

[a] Reaction conditions: 0.2 mmol of (2*E*,4*E*)-**1a,** 2.5 mol% of [Rh], 5 mol% of [Ag], and 5 mol% of (*R*,*R*)-Ph-BPE in EtOH (2.0 mL) at 45 °C for 3 h. [b] Determined by ^{1}H NMR analysis using mesitylene as the internal standard. [c] Determined by HPLC analysis. [d] Reaction conditions: 0.1 mmol of (2*E*,4*E*)-**1a,** 2.5 mol% of [Rh], 5 mol% of [Ag], and 5 mol% of (*R*,*R*)-Ph-BPE in EtOH (1.0 mL) at 45 °C for 3 h.

ee (entry 5), while no desired product was obtained in MeOH and toluene with 70%—74% recovery of **1a** (entries 6 and 7). No desired product **3a** was observed with [Rh(nbd)Cl]$_2$ as the catalyst (entry 8). When $[Rh(C_2H_4)_2Cl]_2$ was used, the yield of **3a** was improved to 36% with an excellent enantioselectivity while the yield of (2*E*,4*E*,9*Z*,11*E*)-**1a'** was 52% (entry 9). Thus, optimal conditions for the molecular recognition have been defined as 2.5 mol% of $[Rh(C_2H_4)_2Cl]_2$, 5 mol% of (*R*,*R*)-Ph-BPE, and 5 mol% of $AgSbF_6$ in EtOH at 45 °C.

With the optimized reaction conditions in hand, the scope of the Rh-catalyzed intramolecular [4+2]-cycloaddition-C=C bond isomerization of allene-dienes was investigated (Table 3). Both R^1 and R^3 may be H or an alkyl group (entries 1—14). This protocol is very general for the R^2 group including 1°-alkyl ranging from C_1 to C_{10} (entries 1—10) and 2°-alkyl group (entry 12). A free hydroxyl group in R^2 could also be well tolerated (entry 13). The absolute configuration of **3k** was established by the X-ray single crystal diffraction study (Figure 1). Since there is no method for the synthesis of the racemic products to determine the ee of the optically active products, (*S*,*S*)-Ph-BPE was used as the ligand to produce the enantiomers of corresponding products with a similar efficiency (right column, Table 3).

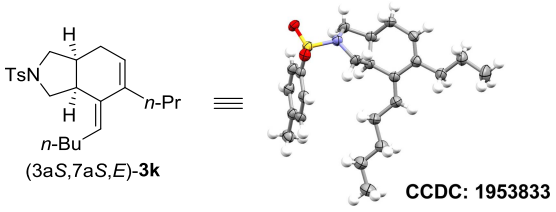
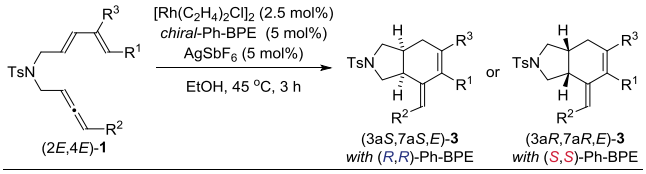


Figure 1 ORTEP representation of (3a*S*,7a*S*,*E*)-**3k**.

To gain insight into the mechanism of isomerization process, the normal [4+2]-cycloaddition product (3a*S*,5*R*,7a*S*,*Z*)-**2a** was prepared according to the known protocol.[7b] No reaction occurred in the absence of the ligand (Scheme 2a, left) while the same C=C isomerization product (3a*S*,7a*S*,*E*)-**3a** was formed in the presence of (*R*,*R*)-Ph-BPE or (*S*,*S*)-Ph-BPE (Scheme 2a, right). Furthermore, when the racemic normal [4+2]-cycloaddition product (3a*R**,5*S**,7a*R**,*Z*)-**2a** was subjected to the standard conditions,

Table 3 Scope study



entry	**1** R^1	R^2	R^3		(3a*S*,7a*S*,*E*)-**3**[a] yield/%	ee/%	(3a*R*,7a*R*,*E*)-**3**[b] yield/%	ee/%
1	Me	Me	H	(**1b**)	34	96	41	98
2	Me	Et	H	(**1c**)	36	>99	36	>99
3	Me	*n*-Pr	H	(**1d**)	30	99	26	98
4	Me	*n*-Bu	H	(**1a**)	37	>99	34[c]	96
5	Me	*n*-C_5H_{11}	H	(**1e**)	34	>99	33	>99
6[d]	Me	*n*-C_6H_{13}	H	(**1f**)	35	>99	35	96
7	Me	*n*-C_7H_{15}	H	(**1g**)	34	>99	35	>99
8	Me	*n*-C_8H_{17}	H	(**1h**)	31	99	33	98
9	Me	*n*-C_9H_{19}	H	(**1i**)	36	>99	32	98
10	Me	*n*-$C_{10}H_{21}$	H	(**1j**)	33	98	31	97
11[e]	*n*-Pr	*n*-Bu	H	(**1k**)	34	98	33	>99
12[f]	Me	Cy	H	(**1l**)	26	97	24	93
13[e,g]	Me	$(CH_2)_7$**OH**	H	(**1m**)	34	>99	33	>99
14[e]	H	*n*-Bu	Me	(**1n**)	40	92	43	92

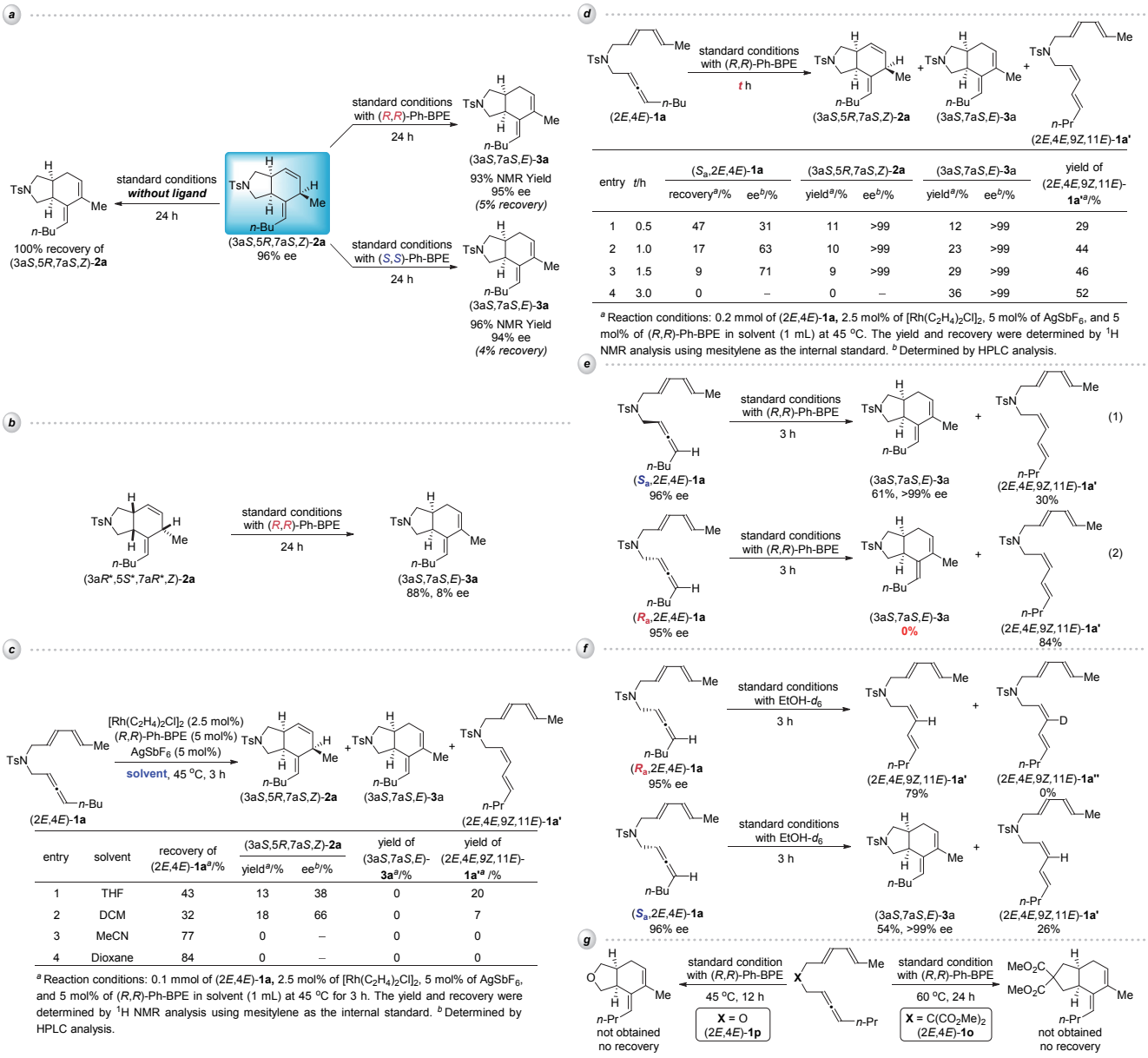
[a] Reaction conditions: 1.0 mmol of (2*E*,4*E*)-**1**, 2.5 mol% of $[Rh(C_2H_4)_2Cl]_2$, 5 mol% of $AgSbF_6$, and 5 mol% of (*R*,*R*)-Ph-BPE in EtOH (5 mL) at 45 °C for 3 h. Isolated yields were presented and ee was determined by HPLC analysis. [b] Reaction conditions: 0.5 mmol of (2*E*,4*E*)-**1**, 2.5 mol% of $[Rh(C_2H_4)_2Cl]_2$, 5 mol% of $AgSbF_6$, and 5 mol% of (*S*,*S*)-Ph-BPE in EtOH (2.5 mL) at 45 °C for 3 h. Isolated yields were presented and ee was determined by HPLC analysis. [c] EtOH (5 mL). [d] Reaction conditions: 0.2 mmol of (2*E*,4*E*)-**1f** was used. [e] 60 °C. [f] 24 h. [g] 12 h.

88% of the C=C bond isomerized product (3a*S*,7a*S*,*E*)-**3a** was obtained with an enantioselectivity of only 8% (Scheme 2b). It clearly revealed that (3a*S*,5*R*,7a*S*,*Z*)-**2a** was formed *via* enantioselective [4+2]-cycloaddition first, which was followed by C=C bond isomerization to afford the final product (3a*S*,7a*S*,*E*)-**3a**. Such C=C bond isomerization may be catalyzed by the *in situ* generated Rh-H species, which was supported by the fact that only the normal [4+2]-cycloaddition product (3a*S*,5*R*,7a*S*,*Z*)-**2a** was generated in a low ee together with the formation of (2*E*,4*E*,9*Z*,11*E*)-**1a'** when running the reaction in aprotic solvent such as THF (tetrahydrofuran) and DCM (dichloromethane) due to the lack of Rh-H species (Scheme 2c, entries 1 & 2). No reaction was observed in MeCN or dioxane (Scheme 2c, entries 3 & 4). To know more about the mechanism, the reaction of (2*E*,4*E*)-**1a** under the standard conditions was monitored by ^{1}H NMR and HPLC analysis (Scheme 2d) and the ees of normal [4+2]-cycloaddition product (3a*S*,5*R*,7a*S*,*Z*)-**2a** and (3a*S*,7a*S*,*E*)-**3a** kept constant while the ee of recovered (S_a,2*E*,4*E*)-**1a** increased, indicating that the normal [4+2]-cycloaddition product (3a*S*,5*R*,7a*S*,*Z*)-**2a** would isomerize to (3a*S*,7a*S*,*E*)-**3a** during the reaction. We also prepared both enantiomers of **1a** *via* EATA (enantioselective allenation of terminal alkynes)[8] and submitted them to the standard conditions, respectively: (3a*S*,7a*S*,*E*)-**3a** was obtained from (S_a,2*E*,4*E*)-**1a** in 61% yield with >99% ee accompanied with 30% yield of (2*E*,4*E*,9*Z*,11*E*)-**1a'**; when (R_a,2*E*,4*E*)-**1a** was applied, (2*E*,4*E*,9*Z*,11*E*)-**1a'** was formed exclusively in 84% isolated yield (Scheme 2e). These results supported that the reaction involves a parallel kinetic resolution process. When (R_a,2*E*,4*E*)-**1a** or (S_a,2*E*,4*E*)-**1a** was subjected to the standard conditions with EtOH-d_6 as solvent, the incorporation of the deuterium atom in product (2*E*,4*E*,9*Z*,11*E*)-**1a''** was not observed, indicating that the isomerization of allene-diene (2*E*,4*E*)-**1** to (2*E*,4*E*,9*Z*,11*E*)-**1'** may not involve Rh-H

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Scheme 2 Mechanistic studies



entry	t/h	(S_a,2*E*,4*E*)-**1a** recovery[a]/%	ee[b]/%	(3a*S*,5*R*,7a*S*,*Z*)-**2a** yield[a]/%	ee[b]/%	(3a*S*,7a*S*,*E*)-**3a** yield[a]/%	ee[b]/%	yield of (2*E*,4*E*,9*Z*,11*E*)-**1a'**[a]/%
1	0.5	47	31	11	>99	12	>99	29
2	1.0	17	63	10	>99	23	>99	44
3	1.5	9	71	9	>99	29	>99	46
4	3.0	0	–	0	–	36	>99	52

[a] Reaction conditions: 0.2 mmol of (2*E*,4*E*)-**1a**, 2.5 mol% of $[Rh(C_2H_4)_2Cl]_2$, 5 mol% of $AgSbF_6$, and 5 mol% of (*R*,*R*)-Ph-BPE in solvent (1 mL) at 45 °C. The yield and recovery were determined by ^1H NMR analysis using mesitylene as the internal standard. [b] Determined by HPLC analysis.

entry	solvent	recovery of (2*E*,4*E*)-**1a**[a]/%	(3a*S*,5*R*,7a*S*,*Z*)-**2a** yield[a]/%	ee[b]/%	yield of (3a*S*,7a*S*,*E*)-**3a**[a]/%	yield of (2*E*,4*E*,9*Z*,11*E*)-**1a'**[a] /%
1	THF	43	13	38	0	20
2	DCM	32	18	66	0	7
3	MeCN	77	0	–	0	0
4	Dioxane	84	0	–	0	0

[a] Reaction conditions: 0.1 mmol of (2*E*,4*E*)-**1a**, 2.5 mol% of $[Rh(C_2H_4)_2Cl]_2$, 5 mol% of $AgSbF_6$, and 5 mol% of (*R*,*R*)-Ph-BPE in solvent (1 mL) at 45 °C for 3 h. The yield and recovery were determined by ^1H NMR analysis using mesitylene as the internal standard. [b] Determined by HPLC analysis.

species (Scheme 2f). We also examined the substrates with other tethers (Scheme 2g). However, the reaction of malonate-tethered allene-diene (2*E*,4*E*)-**1o** and oxygen-tethered (2*E*,4*E*)-**1p** failed to afford the expected products, indicating that the nitrogen atom in the tether has a critical effect on the reaction. We speculated that it may be due to the coordination between the nitrogen atom and rhodium. But the real reason is still unclear.

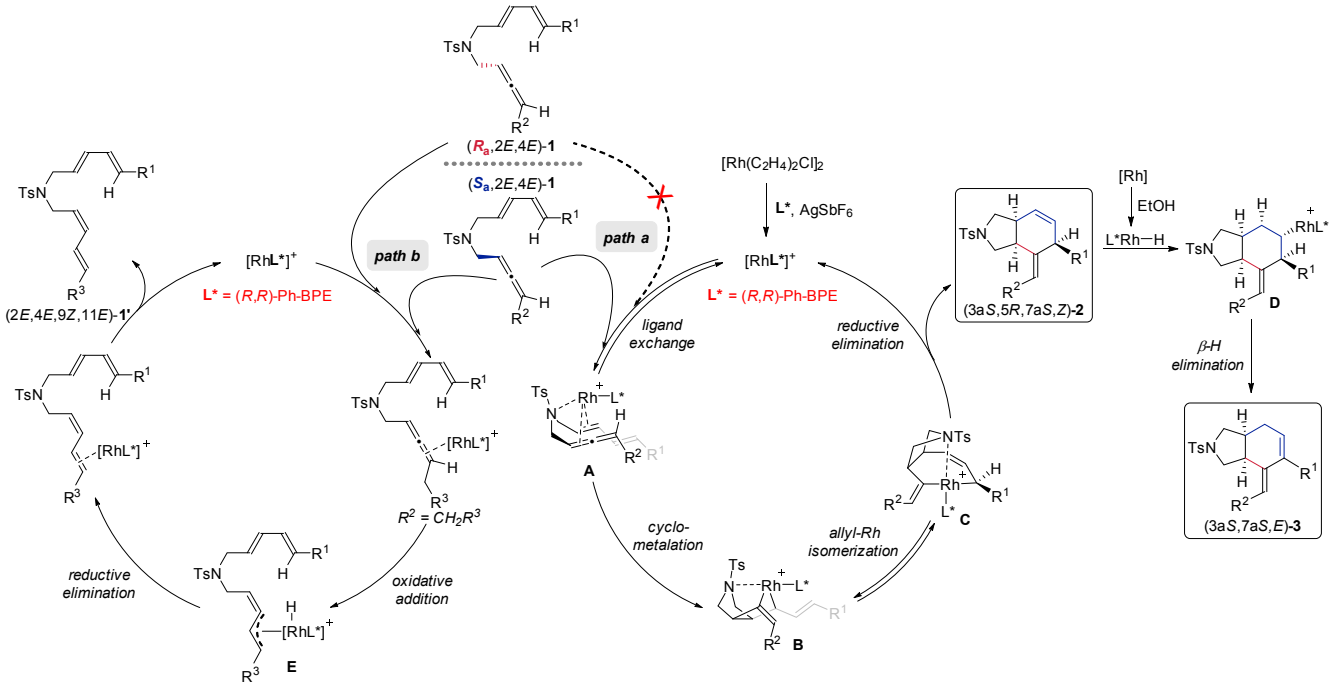
Based on these studies, a possible mechanism is proposed as shown in Scheme 3. Firstly, $[Rh(C_2H_4)_2Cl]_2$ reacted with $AgSbF_6$ and chiral ligand (**L***) to generate chiral cationic catalyst $[RhL^*]^+$, which preferred to react with (S_a,2*E*,4*E*)-**1** to form the intermediate **A** *via* ligand exchange when the chiral ligand was (*R*,*R*)-Ph-BPE (path a). Subsequent cyclometalation afforded intermediate **B**, which would lead to the formation of rhodiabicycloheptene **C** *via* allyl-Rh isomerization. Subsequent reductive elimination yielded the normal [4+2]-cycloaddition product (3a*S*,5*R*,7a*S*,*Z*)-**2** with the regeneration of the catalytically active $[RhL^*]^+$ species. The *in situ* formed Rh hydride species catalyzed the C=C bond isomerization to afford the final product (3a*S*,7a*S*,*E*)-**3** *via* sequential hydrometalation and β-H elimination. (S_a,2*E*,4*E*)-**1** may also slowly coordinate with $[RhL^*]^+$ species and subsequent oxidative addition would afford an allyl-Rh-hydride intermediate **E**, which could yield the product (2*E*,4*E*,9*Z*,11*E*)-**1'** after reductive elimination and dissociation.[9] According to the results shown in Scheme 2e, (R_a,2*E*,4*E*)-**1** isomerized exclusively to (2*E*,4*E*,9*Z*,11*E*)-**1'** *via* path b with (*R*,*R*)-Ph-BPE as the chiral ligand.

The highly functionalized nature of the products makes them very useful in organic synthesis: The reaction of (3a*S*,7a*S*,*E*)-**3** with *m*CPBA (3-chloroperoxybenzoic acid) afforded the double epoxidation product **4** with an excellent diastereoselectivity (Scheme 4a). The absolute configurations of the chiral centers in **4** were established *via* its X-ray diffraction study. In the presence of OsO_4, (3a*S*,7a*S*,*E*)-**3a** may be converted to regioisomeric dihydroxylated products **5a** and **6a**. The oxidation of **6a** in this mixture with Dess-Martin periodinane afforded α-hydroxyl ketone **7a** with the **5a** unreacted (Scheme 4b, up). In addition, when the dihydroxylation mixture was treated with 10 mol% of TsOH in acetone, the diol in **5a** could be protected highly selectively by its treatment with acetone to afford the corresponding ketal **8a** while **6a** was converted to the α,β-unsaturated ketone **9a** (Scheme 4b, down).

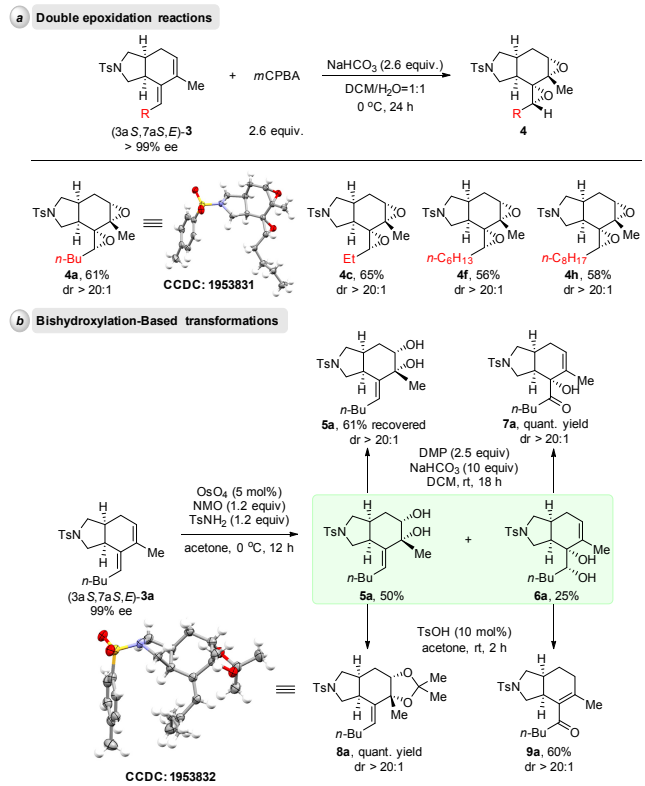
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Scheme 3 Proposed mechanism



Scheme 4 Synthetic applications



Through these processes, highly functionalized optically active pure *cis*-fused bicyclic products **5a**, **7a**, **8a**, and **9a** were prepared efficiently.

Conclusions

In conclusion, we have developed the $[Rh(C_2H_4)_2Cl]_2$/$AgSbF_6$-cocatalyzed kinetic resolution-based [4+2]-cycloaddition of racemic 1,3-disubstituted allene-dienes, which was followed by an exclusive cyclic C=C bond isomerization to afford *cis*-[4.3.0]bicyclic products with an excellent enantio- and *E*-selectivity while the allene unit in the (*R*)-enantiomer has been converted to a conjugated 1,3-diene with (*R*,*R*)-Ph-BPE as the ligand. The bicyclic products have been easily transformed to other useful optically active highly functionalized *cis*-fused bicyclic molecules. The nature of the kinetic resolution process has been carefully unveiled. Further studies on pursuing the kinetic resolution of racemic allene-1,3-dienes are actively ongoing in our laboratory.

Experimental

Asymmetric synthesis of (3a*S*,7a*S*,*E*)-**3a**: To an oven-dried Schlenk tube were added $[Rh(C_2H_4)_2Cl]_2$ (9.8 mg, 0.025 mmol, 99%), (*R*,*R*)-Ph-BPE (25.4 mg, 0.05 mmol), and $AgSbF_6$ (17.1 mg, 0.05 mmol) in a glove box. After replacing air with argon for three times at rt under vacuum, 2.5 mL of EtOH were added. The resulting mixture was stirred at room temperature for 10 min. To the resulting mixture were added (2*E*,4*E*)-**1a** (359.6 mg, 1.0 mmol) and EtOH (2.5 mL) sequentially. The tube was put in an oil bath preheated at 45 °C and the resulting mixture was stirred for 3 h as monitored by TLC. After cooling to rt, the reaction mixture was filtrated through a short pad of silica gel eluted with ethyl acetate (60 mL). After concentration, the residue was purified by chromatography on silica gel to afford (3a*S*,7a*S*,*E*)-**3a** (133.1 mg, 37%) [eluent: petroleum ether/ethyl acetate = 40/1 (500 mL) to 20/1 (300 mL)] as an oil: > 99% ee (HPLC conditions: Chiralcel AS-H column, hexane/*i*-PrOH = 80/20, 1.5 mL/min, λ = 214 nm, t_R (major) = 8.2 min); $[\alpha]_D^{26}$ = +142.0 (*c* = 0.86, $CHCl_3$); ^{1}H NMR (400 MHz, $CDCl_3$) δ: 7.73 (d, *J* = 8.0 Hz, 2 H, ArH), 7.32 (d, *J* = 8.0 Hz, 2 H, ArH), 5.49 (t, *J* = 7.2 Hz, 1 H, =CH), 5.38 (d, *J* = 4.8 Hz, 1 H, =CH), 3.57 (t, *J* = 8.8 Hz, 1 H, one proton of NCH_2), 3.40 (dd, J_1 = 10.0 Hz, J_2 = 5.2 Hz, 1 H, one proton of NCH_2), 3.18 (d, *J* = 10.0 Hz, 1 H, one proton of NCH_2), 3.04—2.92 (m, 1 H, CH), 2.89—2.75 (m, 1 H, one proton of NCH_2), 2.44 (s, 3 H, CH_3), 2.39—2.05 (m, 1 H, one proton of CH_2), 2.09—1.90 (m, 3 H, CH_2 and one proton of CH_2), 1.74 (s, 3 H, CH_3), 1.58—1.43 (m, 1 H, CH), 1.38—1.21 (m, 4 H, 2 x CH_2), 0.91 (t,

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J = 6.8 Hz, 3 H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 143.2, 134.3, 131.7, 131.5, 129.6, 128.1, 127.3, 123.1, 54.0, 49.6, 38.6, 35.7, 32.0, 27.5, 26.0, 22.5, 21.5, 19.8, 13.9; IR (neat) *v*: 2955, 2927, 2859, 1597, 1304, 1180, 1093, 1047 cm^{-1}; MS (70 eV, EI) *m*/*z* (%): 360 (M^++1, 1.26), 359 (M^+, 3.45), 105 (100); HRMS calcd *m*/*z* for $C_{21}H_{29}NO_2S$ (M^+): 359.1914, found: 359.1912.

Supporting Information

The supporting information for this article is available on the WWW under https://doi.org/10.1002/cjoc.202000659.

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